



Data Analyst

Case Study

Jakarta Air Pollution Media Intelligence

Based on real media monitoring data | 4,113 articles | January to November 2023

Role

Data Analyst

Division

Analytics & Insights

Deadline

3 calendar days

What these tests: Can you work with a real, messy dataset, clean it, find the story inside it, and turn that story into a recommendation a client can act on? That is exactly what analysts do at Maverick on Day 1.

THE BRIEF

Your Client & Your Task

Your client is a Jakarta-based startup that makes a mobile air quality reader, a pocket-sized device that measures real-time PM2.5 and AQI levels. The product is good. But almost nobody knows it exists. The timing feels right: Jakarta's air pollution has been all over the news. But the client does not know if this media wave is still an opportunity, or if it has already passed. They have hired Maverick to find out, and it is your job to do the analysis.

THE DATASET

What You Have to Work With

You have been given a raw media monitoring export, 4,113 news articles about air pollution in Indonesia, published between January and November 2023. The file is an Excel spreadsheet with the following columns:

Column	What it contains
Published	Date and time the article was published
Headline	Full article headline (in Indonesian)
Website	The news source, e.g., kompas.com, liputan6.com
Twitter Influencer Shares	How many Twitter accounts with 1K+ followers shared it
Total Facebook	Total Facebook interactions (likes + shares + comments)
Total Interactions	Grand total of all social interactions

Before you start, three data quality notes

- LinkedIn shows zero interactions across all 4,113 rows. This is a tracking error, not a real insight, do not use it in your analysis.
- 482 articles (11.7%) have zero total interactions. This could mean they were not picked up socially, or simply were not tracked by the tool.
- There are no sentiment scores in the data. You will need to read headlines to understand tone.

YOUR TASKS

Three Tasks to Complete

Work through the tasks in order, each one builds on the previous. Your final recommendation in Task 3 should be grounded in what you find in Tasks 1 and 2. Every claim you make must trace back to the data.

Tool tip: You can use Excel, Google Sheets, Python, R, or any other tool you are comfortable with. If you write code, include it as an appendix or a GitHub link.

01

Coverage Volume & Trend

Map when air pollution coverage spiked, how big the wave was, and when it faded. Use this to tell the client whether the media moment is still live, or already over.

What to do:

1. **Monthly volume chart.** Plot how many articles were published each month, January through November 2023. Which month dominates? By how much compared to the average of all other months?
2. **August deep-dive.** Break August into daily counts. On which date did volume first cross 100 articles per day? When did it drop back below that? How long did the peak last?
3. **The trigger.** August went from under 5 articles per day in early August to over 150 per day by mid-August. Look at the headlines from August 11–14. What real-world event caused this sudden jump?
4. **September wave.** September has 716 articles, a second, smaller spike. Is this a continuation of August, or something different? What narrative do the September headlines focus on?

Numbers to work with: August 2023 = 2,819 articles, which is 68.5% of the entire dataset. The other 10 months combined = 1,294 articles.

02

Engagement & Source Analysis

Volume tells you how much was written. Engagement tells you what people actually responded to. Now find out which outlets and articles really moved the needle, and why.

What to do:

1. **Engagement distribution.** Group all articles into these tiers and count how many fall in each: 0 interactions, 1–10, 11–100, 101–1,000, and over 1,000. What does the shape of this distribution tell you?

2. **The outlier.** One article has 25,536 interactions, far more than any other. Find it. What is it about? Why did it perform so much better than everything else? What does that tell you about what Jakarta audiences respond to?
3. **Facebook vs Twitter Influencer Shares.** These two columns measure very different things. Find 2–3 sources that score high on Twitter Influencer Shares but low on Facebook, and 2–3 that are the opposite. What explains the difference?
4. **Source ranking.** Which 5 sources have the highest average engagement per article? (Only count sources with at least 20 articles published.) Compare this list to the top 5 by volume. Are they the same?

Use this reference table as a starting point for your engagement analysis:

Engagement tier	Article count	% of dataset	What this means
> 1,000 interactions	11	0.3%	High impact, mostly political triggers or viral moments
101 – 1,000	97	2.4%	Solid reach, meaningful coverage
11 – 100	1,422	34.6%	Moderate, most 'covered' articles sit here
1 – 10	2,101	51.1%	Minimal, published but barely read socially
0 interactions	482	11.7%	No social pickup, possible tracking gap

Strategic Recommendation

03

Pull everything together. Based on your findings from Tasks 1 and 2, advise the client on when to launch, what to say, and which media to target.

What to do:

1. **Timing.** Should the client launch their awareness campaign now (late 2023), or wait for the next media wave? Use your trend analysis to argue for a specific window. If you think they should wait, what event might trigger the next spike?
2. **Narrative.** Not all coverage angles work for a brand. Look at the types of stories that got the most engagement. Which angle gives the client the most natural entry point, health, lifestyle solutions, celebrity/viral, or something else? Explain your reasoning with data.
3. **Media targets.** Name 3–5 specific outlets the client should approach first. For each one, say why, based on your engagement and platform analysis, not just because they publish a lot.
4. **Data as content.** The client's device produces real-time AQI readings. That is a potential media asset. How could the client use their own data to get journalists interested in covering them? Propose one concrete idea.
5. **AMEC measurement.** Apply the AMEC Integrated Evaluation Framework to your plan. State what outputs you will produce, what change in audience awareness or behaviour you expect (outtakes), and propose 2–3 specific KPIs to measure success in 3 months (outcomes).

What AMEC IEF means in plain terms

Level	Question it answers	Example
Outputs	What did you produce?	5 press releases, 3 media briefings, 1 data report
Outtakes	What did audiences think differently?	Jakarta parents now associate brand X with cleaner air
Outcomes	What changed in the real world?	Brand mentions up 40%, 500 device trial sign-ups

SUBMISSION

How to Submit

Format

There is no required format. Use whatever works best for you: a PowerPoint deck, a PDF report, or a Google Slides presentation. If you write Python or R code as part of your analysis, include it as an appendix or link to a GitHub repository.

What your submission must include

- At least 2 charts or visualisations that are clearly labelled and can be read without additional explanation
- Written analysis for each task, not just charts and bullet points
- A source note for any external data you use beyond the dataset we provided

One thing to keep in mind: We would rather read a submission that takes a clear position and supports it with data, even if the conclusion is not perfect, than one that describes every chart and never commits to a recommendation. The data has a story. Find it and tell it. Be Creative. Good Luck!